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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 18th March Evening Shift

JEE Main Question Paper with Answer Key

18 Mar 2021	90	22	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	21 questions
Physics	25 questions
Out of syllabus	22 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

Deficiency of vitamin K causes :

(A) Cheilosis

(B) Decrease in blood clotting time

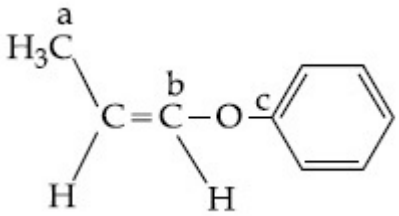
(C) Increase in blood clotting time

(D) Increase in fragility of RBC's

Open solution

Q2. MCQ (Single Correct Answer)

In the following molecule,



Hybridisation of Carbon a, b, and c respectively are :

(A) sp^3, sp^2, sp^2

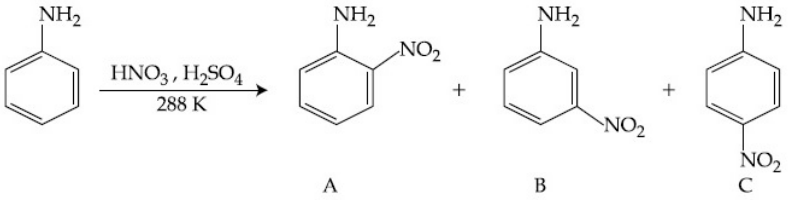
(B) sp^3, sp, sp^2

(C) sp^3, sp^2, sp

(D) sp^3, sp, sp

Open solution

Q3. MCQ (Single Correct Answer)



Consider the given reaction, percentage yield of :

(A) B > C > A

(B) A > C > B

(C) C > A > B

(D) C > B > A

Open solution

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Q4. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : C_2H_5OH and AgCN both can generate nucleophile.

Statement II : KCN and AgCN both will generate nitrile nucleophile with all reaction conditions.

Choose the most appropriate option :

(A) Statement I is false but statement II is true.

(B) Both statement I and statement II are true.

(C) Both statement I and statement II are false.

(D) Statement I is true but statement II is false.

Open solution

Q5. MCQ (Single Correct Answer)

The oxide that shows magnetic property is :

(A) MgO (B) SiO_2 (C) Mn_3O_4 (D) Na_2O

Open solution

Q6. MCQ (Single Correct Answer)

In the reaction of hypobromite with amide, the carbonyl carbon is lost as :

(A) CO_2 (B) CO (C) HCO_3^- (D) CO_3^{2-}

Open solution

Q7. MCQ (Single Correct Answer)

The secondary valency and the number of hydrogen bonded water molecule(s) in $CuSO_4 . 5H_2O$, respectively, are :

(A) 5 and 1 (B) 4 and 1 (C) 6 and 5 (D) 6 and 4

Open solution

Q8. MCQ (Single Correct Answer)

The first ionization energy of magnesium is smaller as compared to that of elements X and Y, but higher than that of Z. The elements X, Y and Z, respectively, are :

(A) neon, sodium and chlorine (B) argon, chlorine and sodium

(C) chlorine, lithium and sodium (D) argon, lithium and sodium

Open solution

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Q9. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Bohr's theory accounts for the stability and line spectrum of Li^+ ion.

Statement II : Bohr's theory was unable to explain the splitting of spectral lines in the presence of a magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but statement II is true.

(B) Both statement I and statement II are true.

(C) Statement I is true but statement II is false.

(D) Both statement I and statement II are false.

Q10. MCQ (Single Correct Answer)

Open solution

The oxidation states of nitrogen in NO, NO_2, N_2O and NO_3^- are in the order of :

(A) $N_2O > NO_2 > NO > NO_3^-$

(B) $NO > NO_2 > N_2O > NO_3^-$

(C) $NO_2 > NO_3^- > NO > N_2O$

(D) $NO_3^- > NO_2 > NO > N_2O$

Q11. MCQ (Single Correct Answer)

Open solution

An organic compound "A" on treatment with benzene sulphonyl chloride gives compound B. B is soluble in dil. NaOH solution. Compound A is :

(A) $C_6H_5-\underset{\displaystyle CH_3}{\underset{|}{CH}}-NH_2$

(B) $C_6H_5 - NHCH_2CH_3$

(C) $C_6H_5 - N - (CH_3)_2$

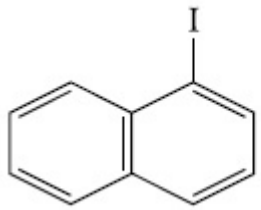
(D) $C_6H_5 - CH_2NHCH_3$

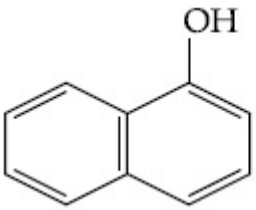
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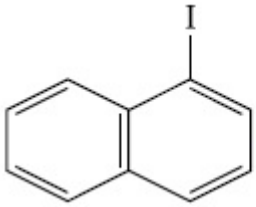
Q12. MCQ (Single Correct Answer)

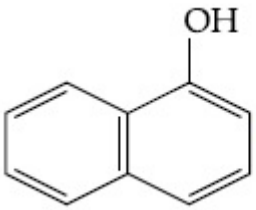
Open solution

Main products formed during a reaction of 1-methoxy naphthalene with hydroiodic acid are :

(A)  and CH_3I

(B)  and CH_3OH

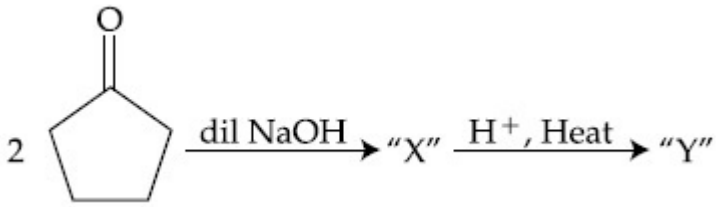
(C)  and CH_3OH

(D)  and CH_3I

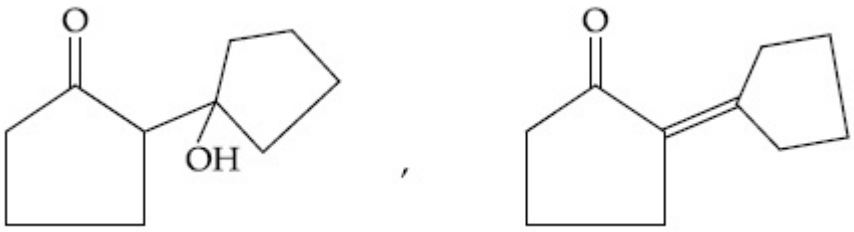
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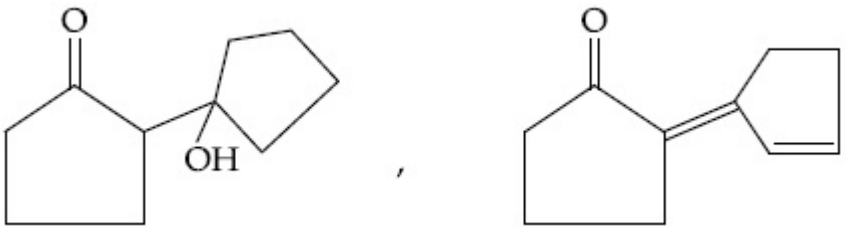
Q13. MCQ (Single Correct Answer)

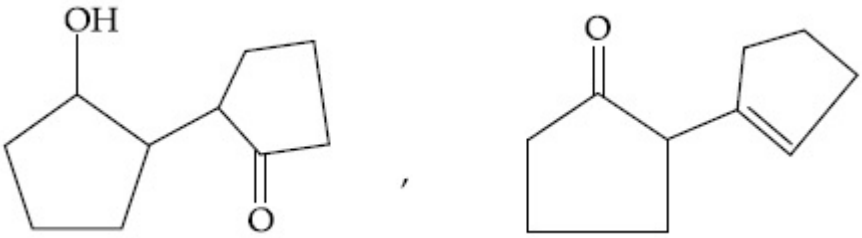
Open solution

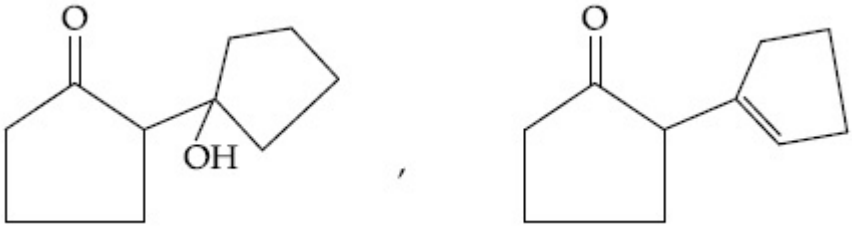


Consider the above reaction, the product 'X' and 'Y' respectively are :

(A) 

(B) 

(C) 

(D) 

Q14. Numerical

Open solution

In Tollen's test for aldehyde, the overall number of electron(s) transferred to the Tollen's reagent formula $[Ag(NH_3)_2]^+$ per aldehyde group to form silver mirror is _____. (Round off to the Nearest Integer).

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Q15. Numerical

10.0 mL of Na_2CO_3 solution is titrated against 0.2 M HCl solution. The following titre values were obtained in 5 readings :

4.8 mL, 4.9 mL, 5.0 mL, 5.0 mL and 5.0 mL

Based on these readings, and convention of titrimetric estimation the concentration of Na_2CO_3 solution is _____ mM.

(Round off to the Nearest Integer).

Open solution

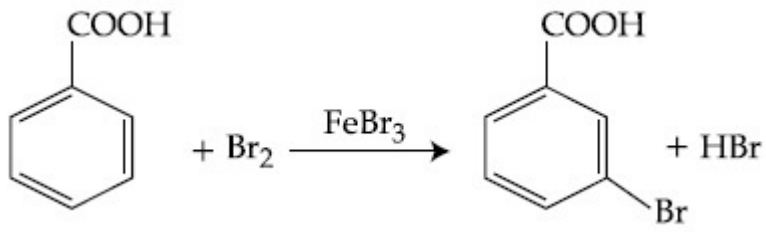
Q16. Numerical

The number of species below that have two lone pairs of electrons in their central atom is _____. (Round off to the Nearest Integer).

SF_4, BF_4^-, ClF_3, AsF_3, PCl_5, BrF_5, XeF_4, SF_6

Open solution

Q17. Numerical



Consider the above reaction where 6.1 g of Benzoic acid is used to get 7.8 g of m-bromo benzoic acid. The percentage yield of the product is _____

(Round off to the Nearest Integer).

[Given : Atomic masses : C : 12.0 u, H : 1.0 u, O : 16.0 u, Br : 80.0 u]

Open solution

Q18. Numerical

The molar conductivities at infinite dilution of barium chloride, sulphuric acid and hydrochloric acid are 280, 860 and 426 S cm^2 mol^{-1} respectively. The molar conductivity at infinite dilution of barium sulphate is _____ S cm^2 mol^{-1}. (Round off to the Nearest Integer).

Open solution

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Q19. Numerical

The gas phase reaction $2A(g) \rightleftharpoons A_2(g)$ at 400 K has ΔG^o = + 25.2 kJ mol^{-1}.

The equilibrium constant K_C for this reaction is _____ $\times 10^{-2}$. (Round off to the Nearest Integer).

[Use : R = 8.3 J mol^{-1} K^{-1}, ln 10 = 2.3, $\log_{10}$ 2 = 0.30, 1 atm = 1 bar]

[antilog (-0.3) = 0.501]

Open solution

Q20. Numerical

A reaction has a half life of 1 min. The time required for 99.9% completion of the reaction is _____ min. (Round off to the Nearest Integer). [Use : ln 2 = 0.69; ln 10 = 2.3]

Open solution

Q21. Numerical

The solubility of $CdSO_4$ in water is 8.0×10^{-4} mol L^{-1}. Its solubility in 0.01 M H_2SO_4 solution is _____ $\times 10^{-6}$ mol L^{-1}. (Round off to the Nearest Integer). (Assume that solubility is much less than 0.01 M)

Open solution

Q22. Numerical

A solute A dimerizes in water. The boiling point of a 2 molal solution of A is 100.52°C. The percentage association of A is _____. (Round off to the Nearest Integer).

[Use : K_b for water = 0.52 K kg mol^{-1} Boiling point of water = 100°C]

Open solution

Mathematics 21 questions

Q23. MCQ (Single Correct Answer)

Let y = y(x) be the solution of the differential equation

$\frac{dy}{dx} = (y+1)\left((y+1)e^{x^2/2} - x\right)$, 0 < x < 2.1, with y(2) = 0. Then the value of $\frac{dy}{dx}$ at x = 1 is equal to :

(A) $\frac{e^{5/2}}{(1+e^2)^2}$

(B) $\frac{5e^{1/2}}{(e^2+1)^2}$

(C) $-\frac{2e^2}{(1+e^2)^2}$

(D) $\frac{-e^{3/2}}{(e^2+1)^2}$

Open solution

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Q24. MCQ (Single Correct Answer)

Let the system of linear equations

$4x + \lambda y + 2z = 0$

$2x - y + z = 0$

$\mu x + 2y + 3z = 0$, $\lambda, \mu \in R$.

has a non-trivial solution. Then which of the following is true?

(A) $\mu = 6, \lambda \in R$ (B) $\lambda = 3, \mu \in R$

(C) $\mu = -6, \lambda \in R$ (D) $\lambda = 2, \mu \in R$

Open solution

Q25. MCQ (Single Correct Answer)

The area bounded by the curve $4y^2 = x^2(4 - x)(x - 2)$ is equal to :

(A) $\frac{\pi}{16}$ (B) $\frac{\pi}{8}$ (C) $\frac{3\pi}{2}$ (D) $\frac{3\pi}{8}$

Open solution

Q26. MCQ (Single Correct Answer)

If $15\sin^4\alpha + 10\cos^4\alpha = 6$, for some $\alpha \in R$, then the value of

$27\sec^6\alpha + 8\operatorname{cosec}^6\alpha$ is equal to :

(A) 500 (B) 400 (C) 250 (D) 350

Open solution

Q27. MCQ (Single Correct Answer)

Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors perpendicular to each other and $|\vec{a}| = |\vec{b}|$. If $|\vec{a} \times \vec{b}| = |\vec{a}|$, then the angle between the vectors $\left(\vec{a} + \vec{b} + \left(\vec{a} \times \vec{b}\right)\right)$ and $\vec{a}$ is equal to :

(A) $\sin^{-1}\left(\frac{1}{\sqrt{6}}\right)$ (B) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(C) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (D) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Open solution

Q28. MCQ (Single Correct Answer)

Let a complex number be $w = 1 - \sqrt{3}i$. Let another complex number z be such that $|zw| = 1$ and $\arg(z) - \arg(w) = \frac{\pi}{2}$. Then the area of the triangle with vertices origin, z and w is equal to :

(A) 4 (B) $\frac{1}{4}$ (C) 2 (D) $\frac{1}{2}$

Open solution

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Q29. MCQ (Single Correct Answer)

Open solution

Let f : R $\rightarrow$ R be a function defined as

$$f(x) = \begin{cases} \frac{\sin(a+1)x+\sin 2x}{2x}, if\, x < 0 \\ b, if\, x = 0 \\ \frac{\sqrt{x+bx^3}-\sqrt{x}}{bx^{5/2}}, if\, x > 0 \end{cases}$$

If f is continuous at x = 0, then the value of a + b is equal to :

(A) -3 (B) -2 (C) $-\frac{5}{2}$ (D) $-\frac{3}{2}$

Q30. MCQ (Single Correct Answer)

Open solution

Let g(x) = $\int_0^x f(t)dt$, where f is continuous function in [0, 3] such that $\frac{1}{3} \leq$ f(t) ≤ 1 for all t$\in$ [0, 1] and $0 \leq$ f(t) $\leq \frac{1}{2}$ for all t$\in$ (1, 3]. The largest possible interval in which g(3) lies is :

(A) $\left[-1, -\frac{1}{2}\right]$ (B) $\left[-\frac{3}{2}, -1\right]$

(C) [1, 3] (D) $\left[\frac{1}{3}, 2\right]$

Q31. MCQ (Single Correct Answer)

Open solution

Let in a series of 2n observations, half of them are equal to a and remaining half are equal to $-$a. Also by adding a constant b in each of these observations, the mean and standard deviation of new set become 5 and 20, respectively. Then the value of $a^2 + b^2$ is equal to :

(A) 425 (B) 250 (C) 925 (D) 650

Q32. MCQ (Single Correct Answer)

Open solution

In a triangle ABC, if $|\overrightarrow{BC}| = 8, |\overrightarrow{CA}| = 7, |\overrightarrow{AB}| = 10$, then the projection of the vector $\overrightarrow{AB}$ on $\overrightarrow{AC}$ is equal to :

(A) $\frac{25}{4}$ (B) $\frac{127}{20}$ (C) $\frac{85}{14}$ (D) $\frac{115}{16}$

Q33. MCQ (Single Correct Answer)

Open solution

Define a relation R over a class of n $\times$ n real matrices A and B as

"ARB iff there exists a non-singular matrix P such that $PAP^{-1} = B$".

Then which of the following is true?

(A) R is reflexive, transitive but not symmetric

(B) R is symmetric, transitive but not reflexive.

(C) R is reflexive, symmetric but not transitive

(D) R is an equivalence relation

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Q34. MCQ (Single Correct Answer)

Let the centroid of an equilateral triangle ABC be at the origin. Let one of the sides of the equilateral triangle be along the straight line x + y = 3. If R and r be the radius of circumcircle and incircle respectively of ΔABC, then (R + r) is equal to :

Open solution

(A) $7\sqrt{2}$ (B) $\frac{9}{\sqrt{2}}$ (C) $2\sqrt{2}$ (D) $3\sqrt{2}$

Q35. MCQ (Single Correct Answer)

Let $f : R - \{3\} \to R - \{1\}$ be defined by $f(x) = \frac{x-2}{x-3}$.

Let $g : R \to R$ be given as $g(x) = 2x - 3$. Then, the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to :

Open solution

(A) 3 (B) 5 (C) 2 (D) 7

Q36. MCQ (Single Correct Answer)

Let S_1 be the sum of first 2n terms of an arithmetic progression. Let S_2 be the sum of first 4n terms of the same arithmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first 6n terms of the arithmetic progression is equal to :

Open solution

(A) 7000 (B) 1000 (C) 3000 (D) 5000

Q37. Numerical

If $\sum_{r=1}^{10} r!(r^3 + 6r^2 + 2r + 5) = \alpha(11!)$, then the value of α is equal to _____.

Open solution

Q38. Numerical

If f(x) and g(x) are two polynomials such that the polynomial $P(x) = f(x^3) + x\, g(x^3)$ is divisible by $x^2 + x + 1$, then P(1) is equal to _____.

Open solution

Q39. Numerical

The term independent of x in the expansion of

$\left[\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right]^{10}$, $x \neq 1$, is equal to _____.

Open solution

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Q40. Numerical

Let y = y(x) be the solution of the differential equation

$xdy - ydx = \sqrt{(x^2 - y^2)}dx$, $x \geq 1$, with y(1) = 0. If the area bounded by the line x = 1, $x = e^{\pi}$, y = 0 and y = y(x) is $\alpha e^{2\pi} + \beta$, then the value of $10(\alpha + \beta)$ is equal to _____.

Open solution

Q41. Numerical

Let I be an identity matrix of order 2×2 and $P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$. Then the value of $n \in N$ for which $P^n = 5I - 8P$ is equal to _____.

Open solution

Q42. Numerical

Let $f : R \to R$ satisfy the equation $f(x + y) = f(x) \cdot f(y)$ for all $x, y \in R$ and $f(x) \neq 0$ for any $x \in R$. If the function f is differentiable at x = 0 and f'(0) = 3, then

$\lim_{h \to 0} \frac{1}{h}(f(h) - 1)$ is equal to _____.

Open solution

Q43. Numerical

Let P(x) be a real polynomial of degree 3 which vanishes at $x = -3$. Let P(x) have local minima at x = 1, local maxima at $x = -1$ and $\int_{-1}^{1} P(x)dx = 18$, then the sum of all the coefficients of the polynomial P(x) is equal to _____.

Open solution

Physics 25 questions

Q44. MCQ (Single Correct Answer)

A plane electromagnetic wave propagating along y-direction can have the following pair of electric field $\left(\vec{E}\right)$ and magnetic field $\left(\vec{B}\right)$ components.

(A) E_x, B_z or E_z, B_x (B) E_x, B_y or E_y, B_x (C) E_y, B_y or E_z, B_z (D) E_y, B_x or E_x, B_y

Open solution

Q45. MCQ (Single Correct Answer)

If the angular velocity of earth's spin is increased such that the bodies at the equator start floating, the duration of the day would be approximately : [Take g = 10 ms^{-2}, the radius of earth, R = 6400×10^3 m, Take π = 3.14]

(A) 84 minutes (B) 1200 minutes (C) 60 minutes (D) does not change

Open solution

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Q46. MCQ (Single Correct Answer)

An object of mass m_1 collides with another object of mass m_2, which is at rest. After the collision the objects move with equal speeds in opposite direction. The ratio of the masses $m_2 : m_1$ is :

(A) 1 : 1 (B) 3 : 1 (C) 2 : 1 (D) 1 : 2

Open solution

Q47. MCQ (Single Correct Answer)

In a series LCR circuit, the inductive reactance (X_L) is 10Ω and the capacitive reactance (X_C) is 4Ω. The resistance (R) in the circuit is 6Ω. The power factor of the circuit is :

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2\sqrt{2}}$

Open solution

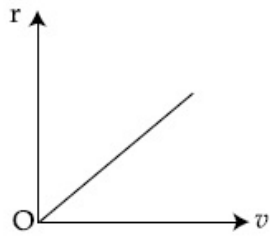
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Q48. MCQ (Single Correct Answer)

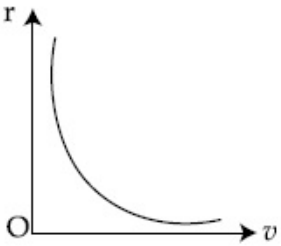
A particle of mass m moves in a circular orbit under the central potential field, $U(r) = -\frac{C}{r}$, where C is a positive constant. The correct radius $-$ velocity graph of the particle's motion is :

Open solution

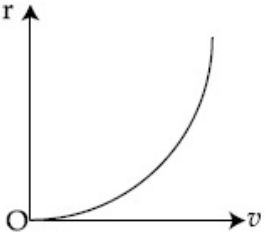
(A)



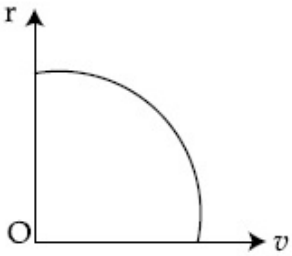
(B)



(C)



(D)



Q49. MCQ (Single Correct Answer)

The angular momentum of a planet of mass M moving around the sun in an elliptical orbit is $\vec{L}$. The magnitude of the areal velocity of the planet is :

Open solution

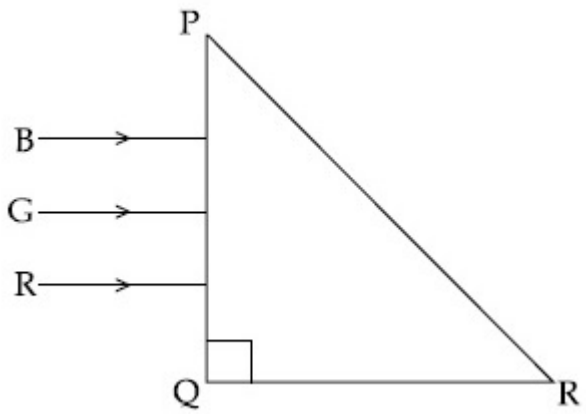
(A) $\frac{2L}{M}$ (B) $\frac{L}{2M}$ (C) $\frac{L}{M}$ (D) $\frac{4L}{M}$

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Q50. MCQ (Single Correct Answer)

Open solution

Three rays of light, namely red (R), green (G) and blue (B) are incident on the face PQ of a right angled prism PQR as shown in the figure.



The refractive indices of the material of the prism for red, green and blue wavelength are 1.27, 1.42 and 1.49 respectively. The colour of the ray(s) emerging out of the face PR is :

(A) green (B) blue and green (C) blue (D) red

Q51. MCQ (Single Correct Answer)

Open solution

Consider a uniform wire of mass M and length L. It is bent into a semicircle. Its moment of inertia about a line perpendicular to the plane of the wire passing through the center is :

(A) $\frac{1}{4}\frac{ML^2}{\pi^2}$ (B) $\frac{1}{2}\frac{ML^2}{\pi^2}$

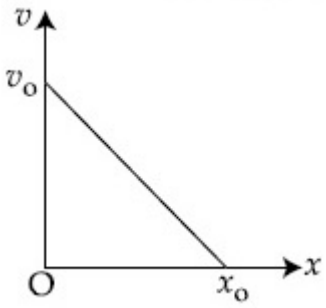
(C) $\frac{2}{5}\frac{ML^2}{\pi^2}$ (D) $\frac{ML^2}{\pi^2}$

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Q52. MCQ (Single Correct Answer)

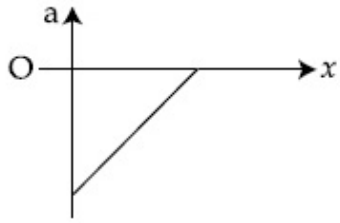
Open solution

The velocity – displacement graph of a particle is shown in the figure.

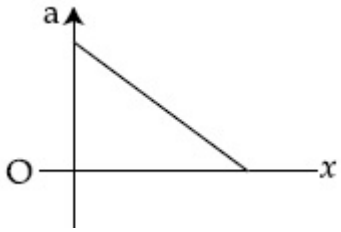


The acceleration – displacement graph of the same particle is represented by :

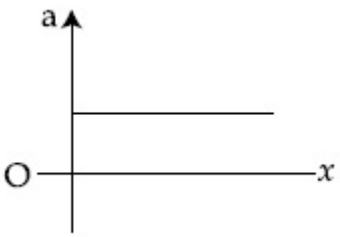
(A)



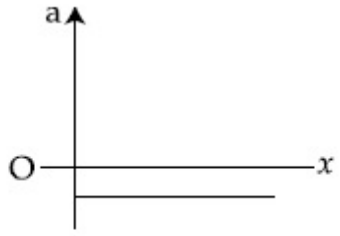
(B)



(C)



(D)



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Q53. MCQ (Single Correct Answer)

Consider a sample of oxygen behaving like an ideal gas. At 300 K, the ratio of root mean square (rms) velocity to the average velocity of gas molecule would be :

(Molecular weight of oxygen is 32g/mol; R = 8.3 J K^{-1} mol^{-1})

(A) $\sqrt{\frac{3\pi}{8}}$ (B) $\sqrt{\frac{3}{3}}$ (C) $\sqrt{\frac{8}{3}}$ (D) $\sqrt{\frac{8\pi}{3}}$

Open solution

Q54. MCQ (Single Correct Answer)

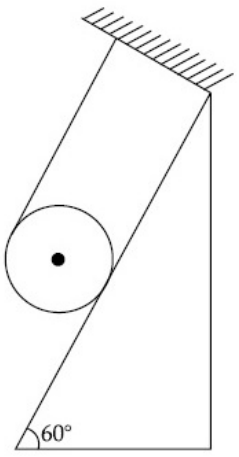
The time taken for the magnetic energy to reach 25% of its maximum value, when a solenoid of resistance R, inductance L is connected to a battery, is :

(A) infinite (B) $\frac{L}{R}$ ln10 (C) $\frac{L}{R}$ ln2 (D) $\frac{L}{R}$ ln5

Open solution

Q55. MCQ (Single Correct Answer)

A solid cylinder of mass m is wrapped with an inextensible light string and, is placed on a rough inclined plane as shown in the figure. The frictional force acting between the cylinder and the inclined plane is :



[The coefficient of static friction, μ_s is 0.4]

(A) 0 (B) 5 mg (C) $\frac{7}{2}$ mg (D) $\frac{mg}{5}$

Open solution

Q56. MCQ (Single Correct Answer)

The function of time representing a simple harmonic motion with a period of $\frac{\pi}{\omega}$ is :

(A) cos(ωt) + cos(2ωt) + cos(3ωt) (B) $\sin^2(\omega t)$

(C) sin(ωt) + cos(ωt) (D) $3\cos\left(\frac{\pi}{4} - 2\omega t\right)$

Open solution

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Q57. MCQ (Single Correct Answer)

For an adiabatic expansion of an ideal gas, the fractional change in its pressure is equal to (where γ is the ratio of specific heats) :

(A) $-\frac{1}{\gamma}\frac{dV}{V}$ (B) $-\gamma\frac{V}{dV}$

(C) $-\gamma\frac{dV}{V}$ (D) $\frac{dV}{V}$

Open solution

Q58. MCQ (Single Correct Answer)

A proton and an α-particle, having kinetic energies K_p and K_α respectively, enter into a magnetic field at right angles.

The ratio of the radii of trajectory of proton to that of α-particle is 2 : 1. The ratio of $K_p : K_\alpha$ is :

(A) 8 : 1 (B) 4 : 1 (C) 1 : 8 (D) 1 : 4

Open solution

Q59. MCQ (Single Correct Answer)

An ideal gas in a cylinder is separated by a piston in such a way that the entropy of one part is S_1 and that of the other part is S_2. Given that $S_1 > S_2$. If the piston is removed then the total entropy of the system will be :

(A) $S_1 - S_2$ (B) $\frac{S_1}{S_2}$ (C) $S_1 \times S_2$ (D) $S_1 + S_2$

Open solution

Q60. MCQ (Single Correct Answer)

Which of the following statements are correct?

(A) Electric monopoles do not exist whereas magnetic monopoles exist.

(B) Magnetic field lines due to a solenoid at its ends and outside cannot be completely straight and confined.

(C) Magnetic field lines are completely confined within a toroid.

(D) Magnetic field lines inside a bar magnet are not parallel.

(E) $\chi = -1$ is the condition for a perfect diamagnetic material, where χ is its magnetic susceptibility.

Choose the correct answer from the options given below :

(A) (A) and (B) only (B) (B) and (C) only (C) (C) and (E) only (D) (B) and (D) only

Open solution

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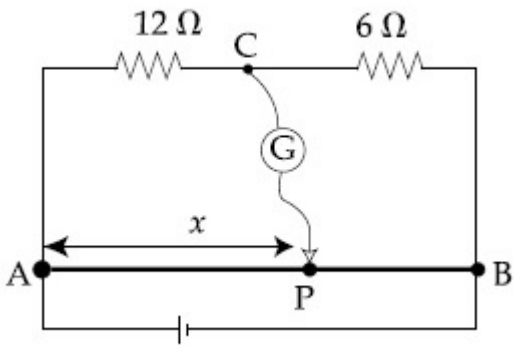
Q61. Numerical

Two wires of same length and thickness having specific resistances 6Ω cm and 3Ω cm respectively are connected in parallel. The effective resistivity is $\rho\Omega$ cm. The value of ρ, to the nearest integer, is _____.

Open solution

Q62. Numerical

Consider a 72 cm long wire AB as shown in the figure. The galvanometer jockey is placed at P on AB at a distance x cm from A. The galvanometer shows zero deflection.



The value of x, to the nearest integer, is _____.

Open solution

Q63. Numerical

A galaxy is moving away from the earth at a speed of 286 kms^{-1}. The shift in the wavelength of a redline at 630 nm is $x \times 10^{-10}$ m. The value of x, to the nearest integer, is _____. [Take the value of speed of light c, as 3×10^8 ms^{-1}]

Open solution

Q64. Numerical

A ball of mass 4 kg, moving with a velocity of 10 ms^{-1}, collides with a spring of length 8 m and force constant 100 Nm^{-1}. The length of the compressed spring is x m. The value of x, to the nearest integer, is _____.

Open solution

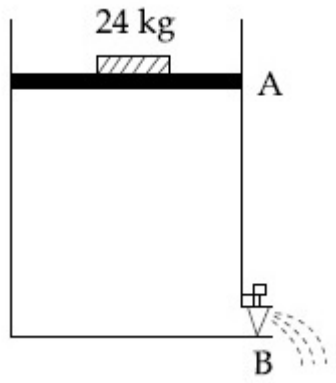
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Q65. Numerical

Open solution

Consider a water tank as shown in the figure. It's cross-sectional area is 0.4 m^2. The tank has an opening B near the bottom whose cross-section area is 1 cm^2. A load of 24 kg is applied on the water at the top when the height of the water level is 40 cm above the bottom, the velocity of water coming out the opening B is v ms^{-1}.

The value of v, to the nearest integer, is _____. [Take value of g to be 10 ms^{-2}]



Q66. Numerical

Open solution

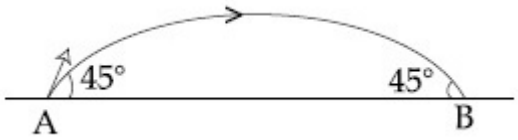
The radius of a sphere is measured to be (7.50 $\pm$ 0.85) cm. Suppose the percentage error in its volume is x.

The value of x, to the nearest integer, is _____.

Q67. Numerical

Open solution

The projectile motion of a particle of mass 5 g is shown in the figure.



The initial velocity of the particle is $5\sqrt{2}$ ms^{-1} and the air resistance is assumed to be negligible. The magnitude of the change in momentum between the points A and B is x $\times$ 10^{-2} $kgms^{-1}$. The value of x, to the nearest integer, is _____.

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Q68. Numerical

An infinite number of point charges, each carrying 1 μC charge, are placed along the y-axis at y = 1 m, 2 m, 4 m, 8 m .....

The total force on a 1C point charge, placed at the origin, is x $\times 10^3$ N.

The value of x, to the nearest integer, is _____. [Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ Nm^2/C^2]

Open solution

Out of syllabus 22 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q69. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements :

Statement I : Non-biodegradable wastes are generated by the thermal power plants.

Statement II : Bio-degradable detergents leads to eutrophication.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but statement II is true.

(B) Both statement I and statement II are true.

(C) Both statement I and statement II are false.

(D) Statement I is true but statement II is false.

Open solution

Q70. MCQ (Single Correct Answer) · **Chemistry**

The charges on the colloidal CdS sol and TiO_2 sol are, respectively :

(A) positive and positive

(B) positive and negative

(C) negative and negative

(D) negative and positive

Open solution

Q71. MCQ (Single Correct Answer) · **Chemistry**

A hard substance melts at high temperature and is an insulator in both solid and in molten state. This solid is most likely to be a/an :

(A) Covalent solid

(B) Molecular solid

(C) Ionic solid

(D) Metallic solid

Open solution

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Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II :

	List - I		List - II
(a)	Be	(i)	treatment of cancer
(b)	Mg	(ii)	extraction of metals
(c)	Ca	(iii)	incendiary bombs and signals
(d)	Ra	(iv)	windows of X-ray tubes
		(v)	bearings for motor engines

Choose the most appropriate answer from the options given below :

(A) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(B) (a)-(iii), (b)-(iv), (c)-(v), (d)-(ii)

(C) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(v)

(D) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

In basic medium, H_2O_2 exhibits which of the following reactions?

(A) $Mn^{2+} \rightarrow Mn^{4+}$

(B) $I_2 \rightarrow I^-$

(C) $PbS \rightarrow PbSO_4$

Choose the most appropriate answer from the options given below :

(A) (B) only

(B) (A), (B) only

(C) (A) only

(D) (A), (C) only

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II :

	List - I		List - II
(a)	Mercury	(i)	Vapour phase refining
(b)	Copper	(ii)	Distillation refining
(c)	Silicon	(iii)	Electrolytic refining
(d)	Nickel	(iv)	Zone refining

Open solution

Choose the most appropriate answer from the option given below :

(A) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(B) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(C) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(D) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

Q75. MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II :

	List - I (Class of Chemicals)		List - II (Example)
(a)	Antifertility drug	(i)	Meprobamate
(b)	Antibiotic	(ii)	Alitame
(c)	Tranquilizer	(iii)	Norethindrone
(d)	Artificial Sweetener	(iv)	Salvarsan

Open solution

Choose the most appropriate match :

(A) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(C) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(D) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

Q76. Numerical · **Chemistry**

A xenon compound 'A' upon partial hydrolysis gives XeO_2F_2. The number of lone pair of electrons present in compound A is _____. (Round off to the Nearest Integer)

Open solution

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Q77. MCQ (Single Correct Answer) · **Mathematics**

A pole stands vertically inside a triangular park ABC. Let the angle of elevation of the top of the pole from each corner of the park be $\frac{\pi}{3}$. If the radius of the circumcircle of ΔABC is 2, then the height of the pole is equal to :

(A) $\frac{1}{\sqrt{3}}$ (B) $2\sqrt{3}$ (C) $\sqrt{3}$ (D) $\frac{2\sqrt{3}}{3}$

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

Let in a Binomial distribution, consisting of 5 independent trials, probabilities of exactly 1 and 2 successes be 0.4096 and 0.2048 respectively. Then the probability of getting exactly 3 successes is equal to :

(A) $\frac{40}{243}$ (B) $\frac{128}{625}$ (C) $\frac{80}{243}$ (D) $\frac{32}{625}$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let $S_1 : x^2 + y^2 = 9$ and $S_2 : (x-2)^2 + y^2 = 1$. Then the locus of center of a variable circle S which touches S_1 internally and S_2 externally always passes through the points :

(A) $\left(\frac{1}{2}, \pm\frac{\sqrt{5}}{2}\right)$ (B) $(1, \pm 2)$

(C) $\left(2, \pm\frac{3}{2}\right)$ (D) $(0, \pm\sqrt{3})$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

Consider a hyperbola $H : x^2 - 2y^2 = 4$. Let the tangent at a point $P(4, \sqrt{6})$ meet the x-axis at Q and latus rectum at $R(x_1, y_1)$, $x_1 > 0$. If F is a focus of H which is nearer to the point P, then the area of ΔQFR is equal to :

(A) $\sqrt{6} - 1$ (B) $\frac{7}{\sqrt{6}} - 2$ (C) $4\sqrt{6} - 1$ (D) $4\sqrt{6}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

If P and Q are two statements, then which of the following compound statements is a tautology?

(A) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow (P \wedge Q)$

(B) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow Q$

(C) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow P$

(D) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow \sim P$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let a tangent be drawn to the ellipse $\frac{x^2}{27} + y^2 = 1$ at $(3\sqrt{3}\cos\theta, \sin\theta)$ where $\theta \in \left(0, \frac{\pi}{2}\right)$. Then the value of θ such that the sum of intercepts on axes made by this tangent is minimum is equal to :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{4}$

Open solution

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Q83. Numerical · **Mathematics**

Let the mirror image of the point (1, 3, a) with respect to the plane $\vec{r}.\left(2\hat{i}-\hat{j}+\hat{k}\right)-b=0$ be $(-3, 5, 2)$. Then, the value of | a + b | is equal to _____.

Open solution

Q84. Numerical · **Mathematics**

Let P be a plane containing the line $\frac{x-1}{3}=\frac{y+6}{4}=\frac{z+5}{2}$ and parallel to the line $\frac{x-1}{4}=\frac{y-2}{-3}=\frac{z+5}{7}$. If the point $(1, -1, \alpha)$ lies on the plane P, then the value of $|5\alpha|$ is equal to _____.

Open solution

Q85. Numerical · **Mathematics**

Let ${}^{n}C_{r}$ denote the binomial coefficient of x^r in the expansion of $(1 + x)^n$. If $\sum_{k=0}^{10}(2^2+3k)^{10}C_k=\alpha.3^{10}+\beta.2^{10}, \alpha, \beta \in R$, then $\alpha + \beta$ is equal to _____.

Open solution

Q86. MCQ (Single Correct Answer) · **Physics**

The correct relation between α (ratio of collector current to emitter current) and β (ratio of collector current to base current) of a transistor is :

(A) $\beta=\frac{\alpha}{1+\alpha}$ (B) $\alpha=\frac{\beta}{1-\alpha}$

(C) $\alpha=\frac{\beta}{1+\beta}$ (D) $\beta=\frac{1}{1-\alpha}$

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

The speed of electrons in a scanning electron microscope is 1×10^7 ms^{-1}. If the protons having the same speed are used instead of electrons, then the resolving power of scanning proton microscope will be changed by a factor of :

(A) $\frac{1}{1837}$ (B) 1837 (C) $\sqrt{1837}$ (D) $\frac{1}{\sqrt{1837}}$

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

The decay of a proton to neutron is :

(A) always possible as it is associated only with β^+ decay

(B) possible only inside the nucleus

(C) not possible as proton mass is less than the neutron mass

(D) not possible but neutron to proton conversation is possible

Open solution

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Q89. Numerical · **Physics**

Open solution

A TV transmission tower antenna is at a height of 20 m. Suppose that the receiving antenna is at :

(i) ground level

(ii) a height of 5 m.

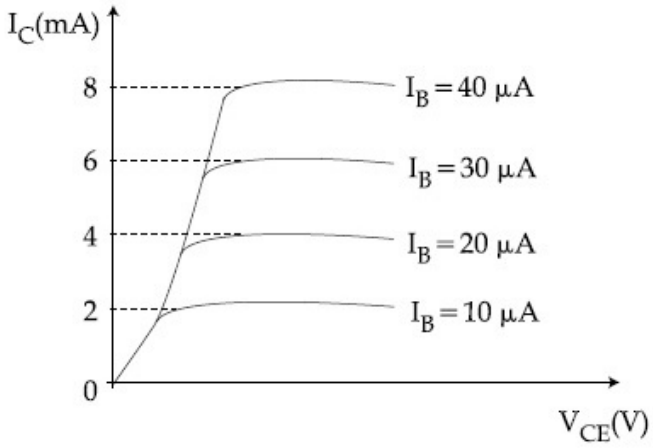
The increase in antenna range in case (ii) relative to case (i) is n%.

The value of n, to the nearest integer, is _____.

Q90. Numerical · **Physics**

Open solution

The typical output characteristics curve for a transistor working in the common-emitter configuration is shown in the figure.



The estimated current gain from the figure is _____.

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Answer Key

Chemistry

1. **C**	2. **A**	3. **D**	4. **D**	5. **C**
6. **D**	7. **B**	8. **B**	9. **A**	10. **D**
11. **A**	12. **D**	13. **A**	14. **2**	15. **50**
16. **2**	17. **78**	18. **288**	19. **2**	20. **10**
21. **64**	22. **100**			

Mathematics

23. **D**	24. **A**	25. **C**	26. **C**	27. **D**
28. **D**	29. **D**	30. **D**	31. **A**	32. **C**
33. **D**	34. **B**	35. **B**	36. **C**	37. **160**
38. **0**	39. **210**	40. **4**	41. **6**	42. **3**
43. **8**				

Physics

44. **A**	45. **A**	46. **B**	47. **C**	48. **B**
49. **B**	50. **D**	51. **D**	52. **A**	53. **A**
54. **C**	55. **D**	56. **D**	57. **C**	58. **B**
59. **D**	60. **C**	61. **4**	62. **48**	63. **6**
64. **6**	65. **3**	66. **34**	67. **5**	68. **12**

Out of syllabus

69. **B**	70. **D**	71. **A**	72. **D**	73. **B**
74. **A**	75. **B**	76. **19**	77. **B**	78. **D**
79. **C**	80. **B**	81. **D**	82. **A**	83. **1**
84. **38**	85. **19**	86. **C**	87. **B**	88. **B**
89. **50**	90. **200**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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